Identity Embedding Run: Status & Cost Report

Prepared: 2026-08-31 · **Scope:** `src/config.py` as committed — `N_SAMPLES=3` x `SAMPLE_SIZE=2000` rows, 16-identity matrix (race x ethnicity x sex x age), raw-CSV mode. Results archived in `data/call_models/0831_identity_embedding_test/`.

Bottom line. All 6 cloud models have completed the identity embedding run — 155–156 of 156 calls each, **935/936 calls, 99.9%**. Estimated spend across the 6 cloud models is ~**$72**, dominated by `claude-sonnet-4-5` (54%) and `gemini-3.1-pro-preview` (23%), the same cost profile as the discovery-phase projection. The 4 local models have **not** been run yet; projected serial runtime is **3–7 hours per model** using discovery-phase latencies. One `gemini-3.1-pro-preview` call is still missing after a transient `503` (not a config/model problem — see §3).

1. Run design

The identity matrix crosses `IDENTITY_RACES` (White, Black or African American) x `IDENTITY_ETHNICITIES` (Hispanic, Not Hispanic) x `IDENTITY_SEXES` (Female, Male) x `IDENTITY_AGES` (35, 55) = **16 identities**. Three of the seven `PROMPT_TYPES` (`identity_prompt`, `identity_hypothetical_prompt`, `emotional_identity_prompt`) are run once per identity; the other four (`control_prompt`, `emotional_prompt`, `emotional_extreme_prompt`, `emotional_suicidal_prompt`) run once, unconditioned. That's **52 (prompt_type, identity) jobs x `N_SAMPLES=3` = 156 calls per model**.

2. Completion status (156 calls/model expected)

Model	Calls done	Missing	Status
`gpt-5-nano`	156/156	0	Complete
`gpt-4o-mini`	156/156	0	Complete
`claude-haiku-4-5`	156/156	0	Complete
`claude-sonnet-4-5`	156/156	0	Complete
`gemini-3.5-flash-lite`	156/156	0	Complete
`gemini-3.1-pro-preview`	155/156	1	1 sample failed on transient `503 UNAVAILABLE`
`qwen2.5-7b-instruct`	0/156	156	Not started
`llama-3.1-8b-instruct`	0/156	156	Not started
`llama-3.2-3b-instruct`	0/156	156	Not started
`gemma-3-12b-it`	0/156	156	Not started

No error entries currently sit in any output file — all API/quota/deprecated-model failures encountered mid-run (`gemini-2.5-*` 404s, one 429 quota, one 503) were stripped as they occurred, so file contents reflect only successful parses.

3. Cloud cost — 156 calls/model, ~82,400 in / 200 out tokens per call

Same token/call estimate as the discovery-phase benchmark (`Discovery_Benchmark_and_Cost_Report.md`) — `SAMPLE_SIZE`/`USE_SUMMARY` are unchanged, so the ~82,401-token raw-CSV prompt still applies. Gemini pricing is carried over from the retired `gemini-2.5-*` tier as a placeholder — ***re-verify `gemini-3.5-flash-lite` / `gemini-3.1-pro-preview` pricing before treating this as final.***

Model	$/1M in · out	$/call	**Run cost (156 calls)**
`claude-sonnet-4-5`	$3.00 · $15.00	$0.2502	$39.03
`gemini-3.1-pro-preview`*	$1.25 · $10.00	$0.1050	$16.38

Model	\$/1M in · out	\$/call	**Run cost (156 calls)**
`claude-haiku-4-5`	\$1.00 · \$5.00	\$0.0834	\$13.01
`gpt-4o-mini`	\$0.15 · \$0.60	\$0.0125	\$1.95
`gemini-3.5-flash-lite`*	\$0.10 · \$0.40	\$0.0083	\$1.30
`gpt-5-nano`	\$0.05 · \$0.40	\$0.0042	\$0.66
Total (cloud, complete)			**~\$72.32**

4. Local models — not yet run

No local-model output exists for this identity run. Projected serial runtime at discovery-phase per-call latency:

Model	Est. runtime (156 calls)
`qwen2.5-7b-instruct`	6.4 h
`gemma-3-12b-it`	5.0 h
`llama-3.1-8b-instruct`	3.4 h
`llama-3.2-3b-instruct`	3.1 h

Carry-forward risk: the discovery benchmark found `llama-3.2-3b-instruct` at 0/10 valid JSON and `llama-3.1-8b-instruct` at 5/10 — unresolved. Running either as-is on this 156-call identity set will likely reproduce the same failure rate; fix formatting/add a repair step before running, or drop the model.

5. Data-quality notes

- **38 empty leftover files** remain in `0831_identity_embedding_test/` under the retired `gemini-2.5-flash-lite` / `gemini-2.5-pro` names (all 0 bytes — their only content was the 404 errors already stripped). Safe to delete; superseded by the `gemini-3.5-flash-lite` / `gemini-3.1-pro-preview` files, which hold the real data.
- **1 missing sample** (`identity_hypothetical_prompt`, `black_nonhispanic_female_age35`, `gemini-3.1-pro-preview`, `sample_0001`) — dropped for a transient `503`, not re-run yet.

6. Simulation: `SAMPLE_SIZE=300` (vs. current 2,000)

Run isolated in scratch — 3 samples (matching current `N_SAMPLES=3`), same seeds as production (`SAMPLE_SEED=42`, seeds 42/43/44), drawn from the same 53,202 eligible rows, labeled with the same regression as `label_samples.py`. No production data was modified for this test.

Label balance. All 3 samples came back `NO_BIAS` (`bias_any` 0/3), and one of the three failed to converge on the logistic regression, falling back to marginal Fisher tests — consistent with the existing calibration curve (`calibration_label_rates.csv`), which shows label positivity dropping sharply below 1,000 rows:

	X=300 (n=3, this sim)	X=1000 (n=150, calib.)	X=2000 (n=150, current)
`bias_any`	**0/3 (0%)**	37.3%	56.0%
`bias_latino_female`	0/3 (0%)	7.3%	12.7%

This is a 3-draw result, not a stable rate estimate — but at `N_SAMPLES=3`, X=300 risks **zero label variance**, which breaks the pairwise-flip / accuracy analysis the pipeline depends on.

Cost. Full prompt re-tokenized with the actual `get_prompt()` template (`o200k_base`): **~12,465 tokens/call** at X=300 vs. ~82,401 at X=2000 — near-linear scaling, fixed template overhead is only 101 tokens.

Scope	X=2000 cost	X=300 cost
156 calls/model (this run's scale)	~$72.32	**~$11.77**
3,500 calls/model (discovery-doc full-study scale)	~$1,622.62	**~$264.11**

Verdict: X=300 is ~6.6x cheaper but, at only 3 samples, is likely too small to produce usable ground-truth label variance. Cutting cost by raising `N_SAMPLES` at a mid-range `SAMPLE_SIZE` (e.g. keeping 2,000) is probably a better lever than shrinking `SAMPLE_SIZE` down to 300.